

TM Forum Component

Product Order Capture and Validation

TMFC002

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Notice

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Direct inquiries to the TM Forum office:

100 Enterprise Drive
Suite 301 #1649
Rockaway, NJ 070866, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

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TMFC002 – Product Order Capture and Validation

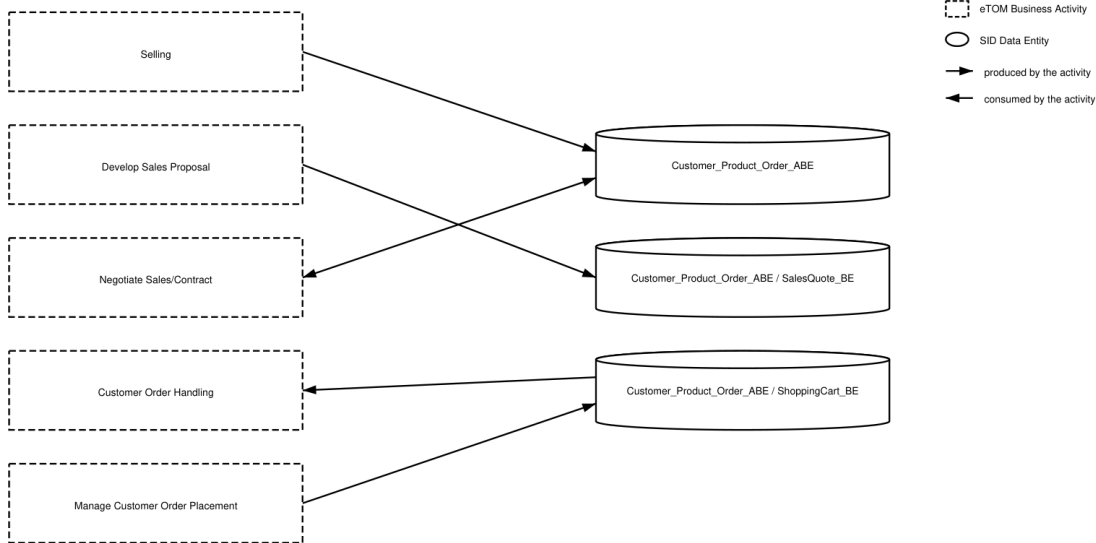
1. Overview

Component Name	ID	Description	ODA Function Block
Product Order Capture and Validation	TMFC002	This component captures what a customer wants to order based on the CSP's Product Catalog. It enables configuration of the product offerings and products desired, provides quotes, checks the eligibility of the customer order, and completes it with information needed such as the related parties or associated billing account and the delivery appointment. This component owns quote management, order capture and validation, using dedicated components (eg Offering Configurator, Service Qualification, Party Management) when needed. After the delivery of the customer product order items, this component is also in charge of the commercial closure of the order. It includes update of the Product Inventory (status and starting/end date of tariffs and discounts) and potential commercial rules control (eg receipt of the contract document signed by the customer).	Core Commerce

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links

TMFC002 eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.1.9	L2	Selling	Responsible for managing prospective customers, for qualifying and educating customers, and matching customer expectations / Managing prospective parties with whom an enterprise may do business, such as potential existing or new customers and partners, for qualifying and educating them, and ensuring their expectations are met.
1.1.9.2	L3	Develop Sales Proposal	Develop a sales proposal to respond to the customer's requirements / Develop a sales proposal to respond to a sales prospect's requirements.
1.1.9.5	L3	Negotiate Sales/Contract	Close the sale with terms that are understood by the customer and are mutually agreeable to both the customer and the service provider. / Close the sale with terms that are understood by the sales prospect, which now becomes a customer or some other party

1.3.3	L2	Customer Order Handling	Responsible for accepting and issuing orders. / Customer Order Processing Management business process directs and controls all activities that operationally realize orders for customer. / Customer Order Processing Management assures the capture, processing, fulfillment, "shipping", delivery and reporting of customer orders from feasibility assessment, purchasing, payment, fulfillment and follow up with the customer for closure.
1.3.3.10	L3	Manage Customer Order Placement	Manage Customer Order Placement business activity directs and controls the capture of information to enable create customer order, change customer order and validate customer order based on ordering information from customer.

2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition
Customer Product Order	Handles single customer orders and the various types thereof, such as regulated and non-regulated orders. / Orders ProductOfferings and concerned Products	SalesQuote	A SalesQuote is a document that allows a Prospect to view the cost involved in purchasing specific Products. It is generally used by suppliers or resellers to begin a business transaction and specifies validity period for acceptance of the quote and possibly an authorisation process depending on the value of the quote.

Customer Product Order	Handles single customer orders and the various types thereof, such as regulated and non-regulated orders. / Orders ProductOfferings and concerned Products	ShoppingCart	Shopping Cart entity is used for the temporarily selection and reservation of product offerings in e-commerce and retail purchase. Shopping cart supports purchase of both tangible and intangible goods and service (e.g. handset, telecom network service). The charge includes the one-off fee such as the fee for handset and the recurring fee such as the fee of a network service. It is mainly used in case of unknown customer with only a ContactMedium instead of a Party and PartyRole with their ContactMedium.
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2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
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16	Fallout Automated Correction	Fallout Management	Fallout Correction Management	Fallout Automated Correction function tries to automatically fix fallouts in workflows before they go to a human for handling. / This includes a Fallout Rules Engine that provides the capability to handling various errors or error types based on built rules. These rules can facilitate autocorrection, correction assistance, placement of errors in the appropriate queues for manual handling, as well as access to various systems.
17	Fallout Correction Information Collection	Fallout Management	Fallout Correction Management	Fallout Correction Information Collection collects relevant information for errors or situations that cannot be handled via Fallout Auto Correction. The intent is to reduce the time required by the technician in diagnosing and fixing the fallout.

18	Fallout Management to Fulfillment Application Accessing	Fallout Management	Fallout Repository Management	Fallout Management to Fulfillment Application Accessing function provides a variety of tools to facilitate Fallout Management access to other applications and repositories to facilitate proper Fallout Management. This can include various general access techniques such as messaging, publish and subscribe, etc. as well as specific APIs and contracts to perform specific queries or updates to various applications or repositories within the fulfillment domain.
19	Fallout Manual Correction Queuing	Fallout Management	Fallout Correction Management	Fallout Manual Correction Queuing function provides the required functionality to place error fallout into appropriate queues to be handled via various staff or workgroups assigned to handle or fix the various types of fallout that occurs during the fulfillment process. This includes the ability to create and configure queues, route errors to the appropriate queues, as well as the ability for staff to access and address the various fallout instances within the queues.

20	Fallout Notification	Fallout Management	Fallout Repository Management	Fallout Notification function provides the means to alert people or workgroups of some fallout situation. This can be done via a number of means, including email, paging, (Fallout management interface bus) etc. This function is done via business rules.
21	Fallout Orchestration	Fallout Management	Fallout Correction Management	The Fallout Orchestration function provides workflow and orchestration capability across Fallout Management.
22	Fallout Reporting	Fallout Management	Fallout Repository Management	Fallout Reporting provides various reports regarding Fallout Management, including statistics on fallout per various times periods (per hour, week, month, etc.) as well as information about specific fallout.
23	Fallout Dashboard System Log-in Accessing	Fallout Management	Fallout Repository Management	Fallout Dashboard System Log-in Accessing provides auto logon capability into various applications needed to analyze and fix fallout

24	Pre-populated Fallout Information Presentation	Fallout Management	Fallout Correction Management	Pre-populated Fallout Information Presentation automatically position the analyzer on appropriate screens pre-populated with information about the order(s) that's subject for fallout handling.
120	Customer Order Capturing	Product Configuration & Activation	Offer and Product Configuration	Customer Order Capturing provides access to Order capture and negotiation capabilities or receives the captured Customer Order data from channels. Takes care of persistence using Customer Order Lifecycle Management. Including support of contract printing, integration with a locally installed cash management/cash register and a retail inventory system for order completion and ordered product versioning.
176	Customer Order Capturing Access	Fulfillment Integration Management	Customer Fulfillment Access Management	Customer Order Capturing Access provides front end support for the Customer Order Capturing functions defined by the order management.

177	Customer Order Take-over Management	Customer Order Management	Customer Order Completion	Customer Order Take-over Management provides an ability to take over governance of orders handled by other channels (e.g., self-service) amend and relinquish while preserving all the captured data.
181	Product Order Data Collection	Customer Order Management	Customer Order Completion	Product Order Data Collection provides an aid in verification and issuance of a complete and valid customer order. This function checks delivery address, link with a payment, a billing account, a certified holder, etc.
204	Customer Order Completion	Customer Order Management	Customer Order Completion	The Customer Order Entry Finalization function enables finalization of the Customer Order with collection of Customer Data or installed base data according to the Catalog. It allows to complete the configuration element not necessary for the quotation. The complements could also concern links with actors, dates, address, billing account.

217	Customer Order Establishment Tracking	Customer Order Management	Customer Order Repository Management	Customer Order Establishment Tracking provides the functionality necessary to track and manage the distributed requests decomposed by Customer Order Distribution.
236	Customer Loyalty Subscription Management	Product Configuration & Activation	Offer and Product Configuration	Customer Loyalty Subscription Configuration function manages information for subscription and deactivation to a Loyalty program. / Loyalty Subscription Management can assign multiple traffic channels to loyalty subscriptions (sim-cards, PBX, Call Data Network etc.)
259	External Call Center Access	Customer Order Management	Customer Order Repository Management	External Call Center Access provides access to call center self-empowered fulfillment function providing an internet technology driven interface for the customer to undertake a variety of fulfillment functions directly for themselves.
277	Shopping Cart Purchasing Access	Customer Order Management	Customer Order Repository Management	Shopping Cart Purchasing Access function provide an internet technology driven interface to the customer to undertake self-service purchase.

317	Product Availability Area Checking	(no description available)	(no description available)	(no description available in the source document or the Functional Framework spreadsheets)
343	Mass Transaction Ordering	Product Configuration & Activation	Offer and Product Configuration	Mass Transaction Ordering feed function of new orders. For bulk sales by e.g., affiliates and corporate customer self-empowered fulfillment provided e.g., by an internet technology driven interface for bulk ordering.
379	Product Customization Offering Management	Sales Management	Opportunity Management	Product Customization Offering Management provides the necessary functionality to manage the customer personalized proposals, taking into account the customer location, needs, current products, as well as the service provider's products, sales emphasis and targets, etc.
388	Sales Order Reporting	Sales Reporting	Sales Performance Management	Sales Order Reporting provides reporting of order handoff.

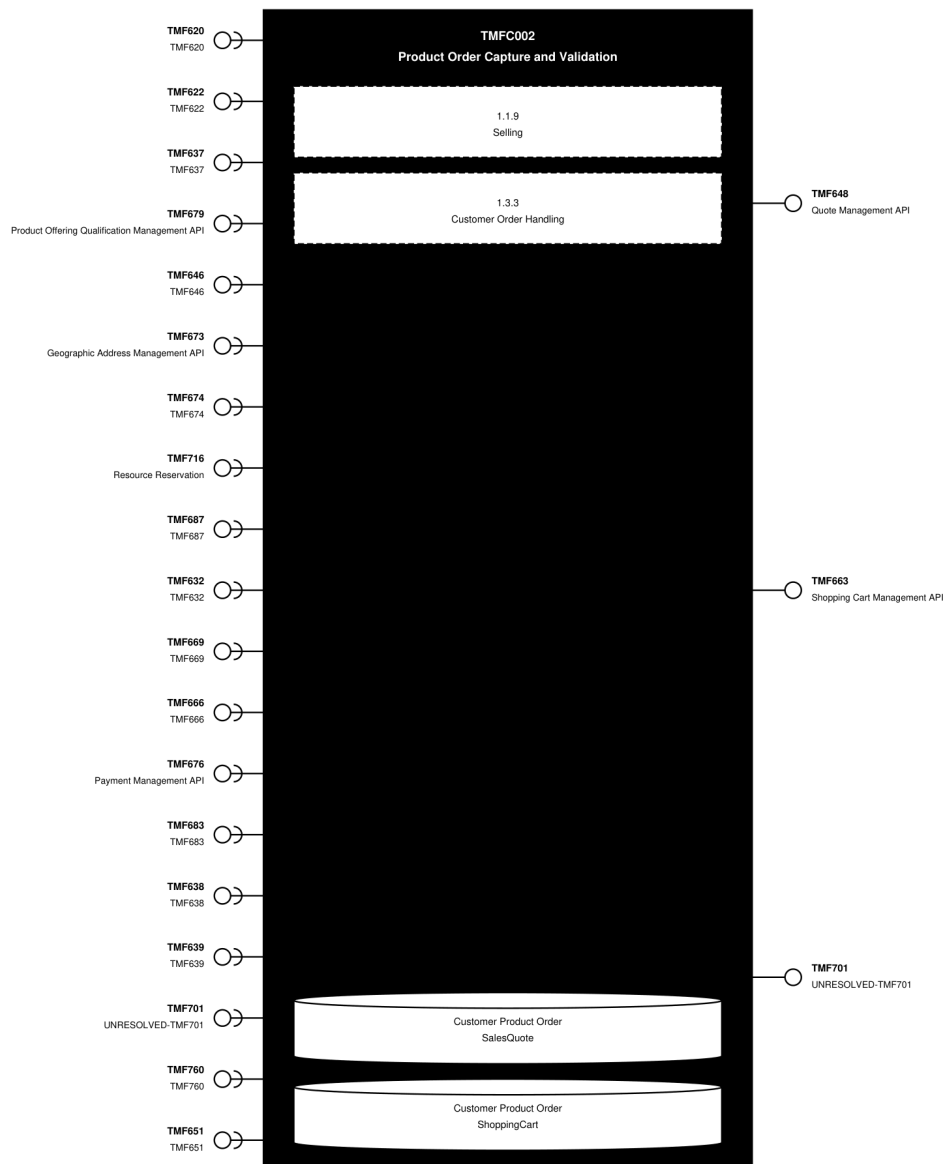
756	Fallout Rule Based Error Correction	Fallout Management	Fallout Correction Management	Fallout Rule Based Error Correction function provides the capability to handle various errors or error types based on pre-defined rules. These rules can facilitate autocorrection, correction assistance, placement of errors in the appropriate queues for manual handling, as well as access to various systems via the Fallout Interface Bus.
934	Sales Negotiation Support	Sales Management	Opportunity Management	Sales Negotiation Support provides support for negotiating the sale by providing multiple quotations as needed, taking into account the customer data, customer qualification, and offers made. Functionality includes access to products, product pricing, scheduling of appointments if a dispatch is necessary, etc. / Sales Negotiation generates A Customer order request. / Customer order request generation – send solution design details to update proposal product order in Customer Order Management systems

1063	Sales Quote Management	Sales Management	Opportunity Management	Sales Quote Management creates and manages quotes, including internal approval.
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3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

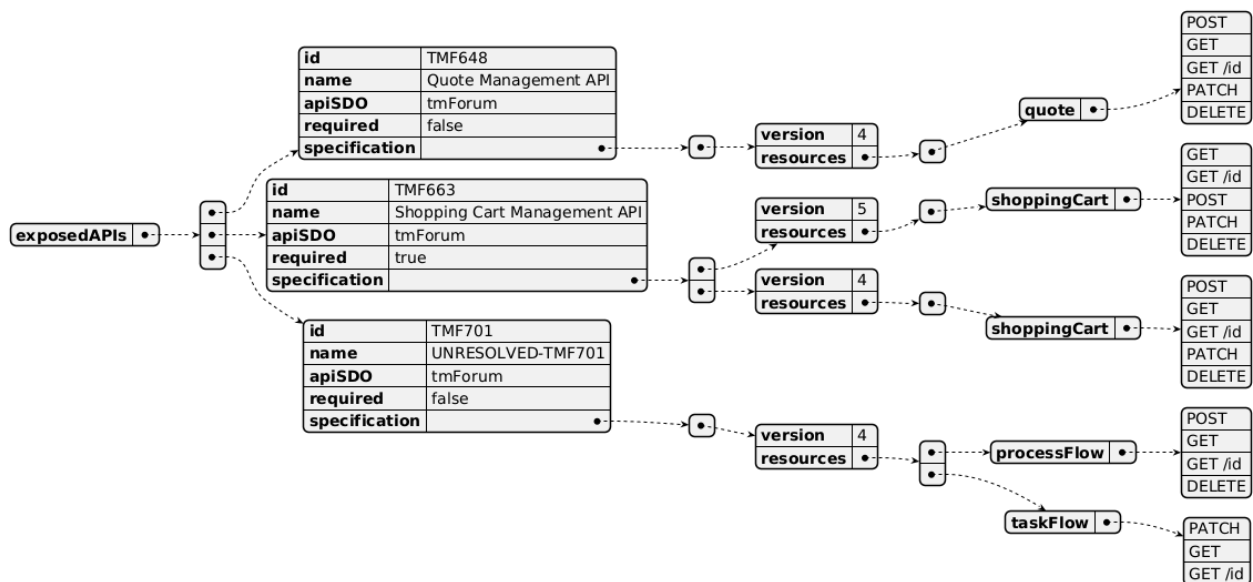


(SVG source: [TMFC002_API_Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the *component-specification-documentation skill's scripts/render_api_context_svg.py*.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations

TMF648	Quote Management API	Optional	4	quote	POST, GET, GET /id, PATCH, DELETE
TMF663	Shopping Cart Management API	Mandatory	5	shoppingCart	GET, GET /id, POST, PATCH, DELETE
TMF663	Shopping Cart Management API	Mandatory	4	shoppingCart	POST, GET, GET /id, PATCH, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	PATCH, GET, GET /id



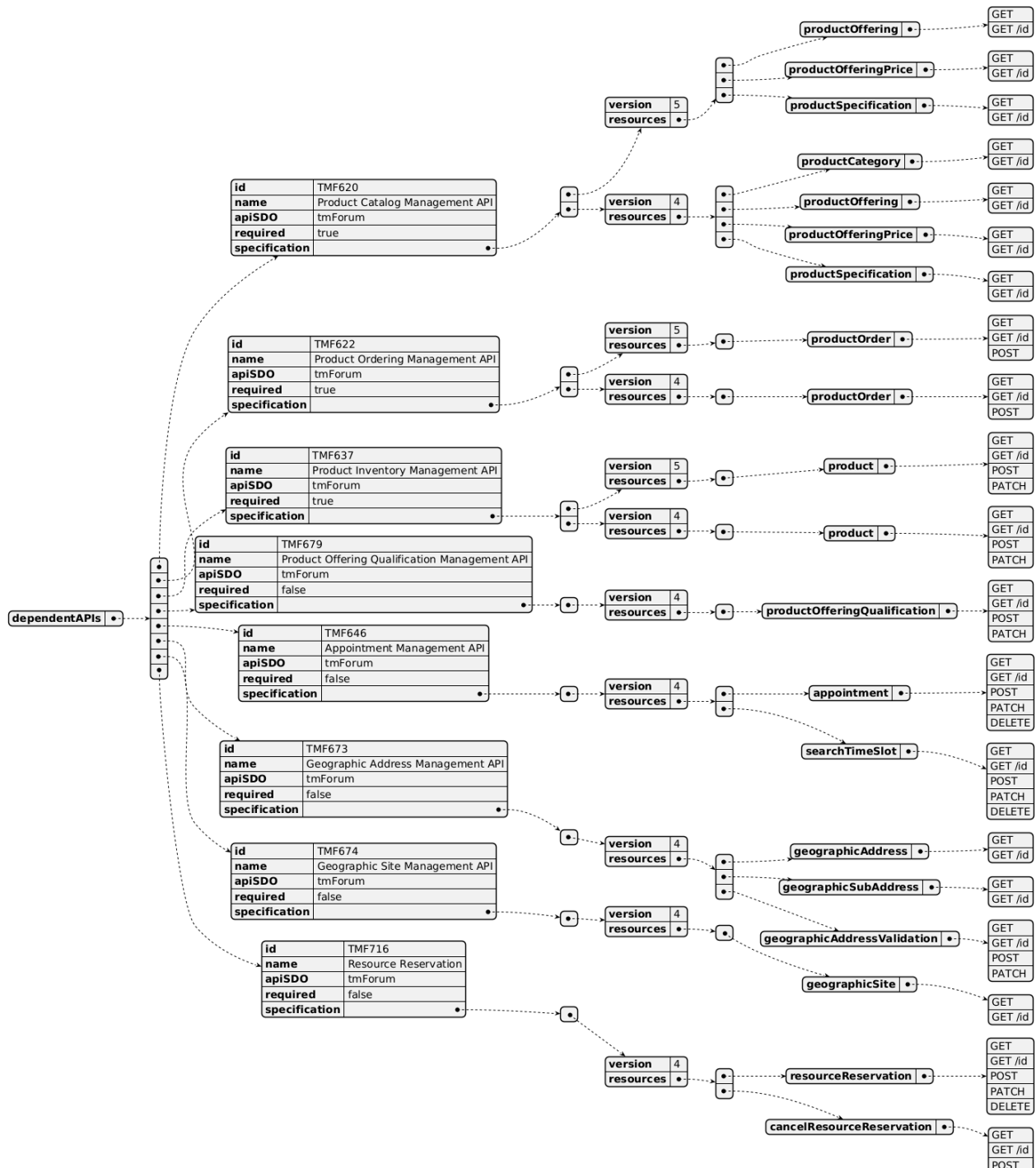
3.3. Dependent APIs

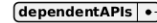
API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF620	TMF620	5	Mandatory	productOffering	GET, GET /id
TMF620	TMF620	5	Mandatory	productOfferingPrice	GET, GET /id
TMF620	TMF620	5	Mandatory	productSpecification	GET, GET /id
TMF620	TMF620	4	Mandatory	productCategory	GET, GET /id
TMF620	TMF620	4	Mandatory	productOffering	GET, GET /id

TMF620	TMF620	4	Mandatory	productOfferingPrice	GET, GET /id
TMF620	TMF620	4	Mandatory	productSpecification	GET, GET /id
TMF622	TMF622	5	Mandatory	productOrder	GET, GET /id, POST
TMF622	TMF622	4	Mandatory	productOrder	GET, GET /id, POST
TMF637	TMF637	5	Mandatory	product	GET, GET /id, POST, PATCH
TMF637	TMF637	4	Mandatory	product	GET, GET /id, POST, PATCH
TMF679	Product Offering Qualification Management API	4	Optional	productOfferingQualification	GET, GET /id, POST, PATCH
TMF646	TMF646	4	Optional	appointment	GET, GET /id, POST, PATCH, DELETE
TMF646	TMF646	4	Optional	searchTimeSlot	GET, GET /id, POST, PATCH, DELETE
TMF673	Geographic Address Management API	4	Optional	geographicAddress	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicSubAddress	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicAddressValidation	GET, GET /id, POST, PATCH
TMF674	TMF674	4	Optional	geographicSite	GET, GET /id
TMF716	Resource Reservation	4	Optional	resourceReservation	GET, GET /id, POST, PATCH, DELETE
TMF716	Resource Reservation	4	Optional	cancelResourceReservation	GET, GET /id, POST

TMF687	TMF687	4	Optional	checkProductStock	GET, GET /id, POST, DELETE
TMF687	TMF687	4	Optional	queryProductStock	GET, GET /id, POST, DELETE
TMF687	TMF687	4	Optional	reserveProductStock	GET, GET /id, POST, DELETE
TMF687	TMF687	4	Optional	productStock	GET, GET /id
TMF632	TMF632	5	Optional	individual	GET, GET /id
TMF632	TMF632	5	Optional	organization	GET, GET /id
TMF632	TMF632	4	Optional	individual	GET, GET /id
TMF632	TMF632	4	Optional	organization	GET, GET /id
TMF669	TMF669	5	Optional	partyRole	GET, GET /id
TMF669	TMF669	4	Optional	partyRole	GET, GET /id
TMF666	TMF666	5	Optional	billingAccount	GET, GET /id
TMF666	TMF666	4	Optional	billingAccount	GET /id, GET
TMF676	Payment Management API	4	Optional	payment	GET /id, GET
TMF683	TMF683	5	Optional	partyInteraction	GET, GET /id
TMF683	TMF683	4	Optional	partyInteraction	GET /id, GET
TMF638	TMF638	5	Optional	service	GET, GET /id
TMF638	TMF638	4	Optional	service	GET /id, GET
TMF639	TMF639	4	Optional	resource	GET /id, GET
TMF701	UNRESOLVED-TMF701	4	Optional	processFlow	POST, PATCH, GET, GET /id
TMF701	UNRESOLVED-TMF701	4	Optional	taskFlow	POST, PATCH, GET, GET /id

TMF760	TMF760	5	Optional	checkProductConfiguration	GET, GET /id, POST
TMF760	TMF760	5	Optional	queryProductConfiguration	GET, GET /id, POST
TMF651	TMF651	4	Optional	agreement	GET, GET /id





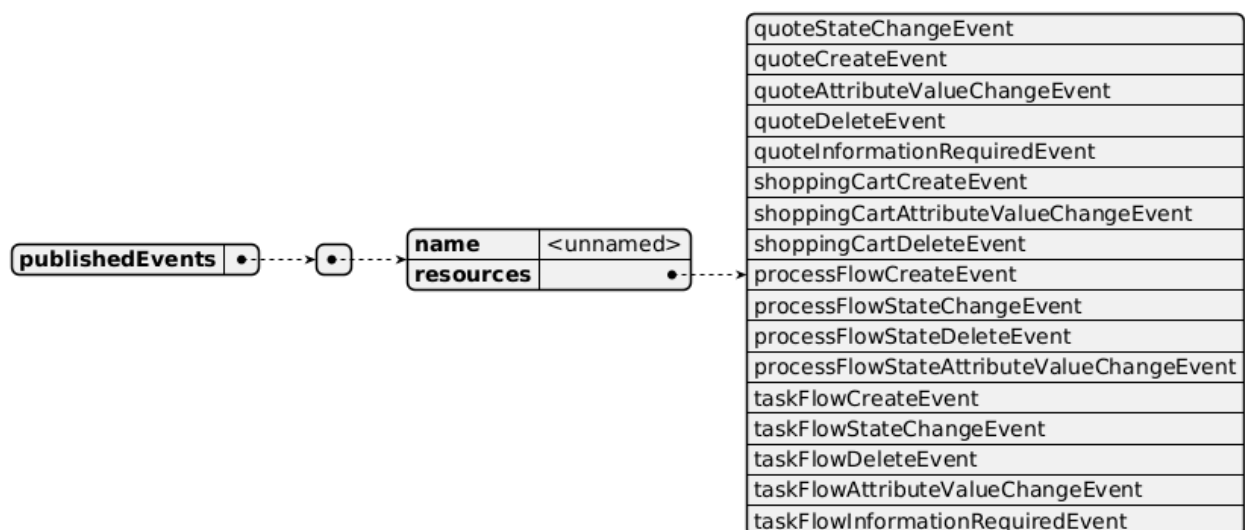
3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in

the preceding sections.

Published Events

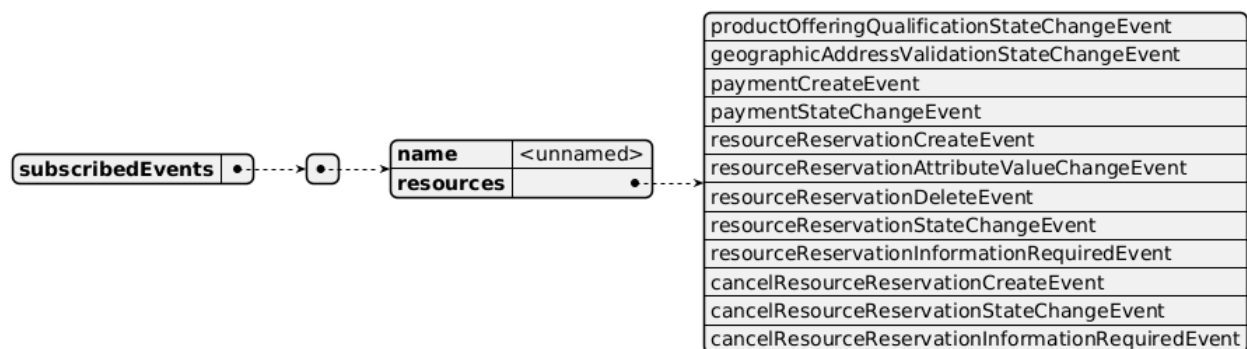
API ID	API Name	Event Resources
TMF648	Quote Management API	quoteStateChangeEvent quoteCreateEvent quoteAttributeValueChangeEvent quoteDeleteEvent quoteInformationRequiredEvent
TMF663	Shopping Cart Management API	shoppingCartCreateEvent shoppingCartAttributeValueChangeEvent shoppingCartDeleteEvent
TMF701	UNRESOLVED-TMF701	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent



Subscribed Events

API ID	API Name	Event Resources
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TMF679	Product Offering Qualification Management API	productOfferingQualificationStateChangeEvent
TMF673	Geographic Address Management API	geographicAddressValidationStateChangeEvent
TMF676	Payment Management API	paymentCreateEvent paymentStateChangeEvent
TMF716	Resource Reservation	resourceReservationCreateEvent resourceReservationAttributeValueChangeEvent resourceReservationDeleteEvent resourceReservationStateChangeEvent resourceReservationInformationRequiredEvent cancelResourceReservationCreateEvent cancelResourceReservationStateChangeEvent cancelResourceReservationInformationRequiredEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v23.0
SID	v25.0

Functional Framework	v23.0
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5.2. Jira References

eTOM - Review L2 1.2.9 - Product Offering Purchasing and clarify L3 definitions and inbound / outbound terminology; delete L3 1.2.9.8 - Report Product Offering Purchase as redundant with 1.3.3.6 - Report Customer Order Handling and 1.6.8.6 - Report Business Partner Orders - Review L3 1.3.3.4 - Complete Customer Order definition as it mixed customer information, customer contracts and service orders. Also review L3 1.3.3.7 - Close Customer Order definition and clarify close/complete terminology and link with 1.3.3.4 - Complete Customer Order (do we need 2 L3?) - 1.3.3.8 - Manage Order Fallout: as this L3 is shared by POCV and POOM components (to be consistent with Functional Framework mapping), clarify which type of fallout needs to be managed during the order delivery orchestration (at POOM level) and which needs to be managed at Product Order Capture and Follow-Up level (by POCV)

SID (*none currently listed*)

Functional Framework These updates need to be studied for R22.0 or R22.5: - Function 204 - Customer Order Completion: rename the function "Customer Order Entry Finalization" as "completion" is confusing here - Function 269 - Installation Preference Configuration: review the definition of this function or delete it (why limited to installation? Why a dedicated function to provide an interface for the customer to capture a set of information?) - Add function(s) to describe the closure step of the customer order - Function 934 - Sales Negotiation Support: review definition and delete part related to service order generation

API - How can we trigger the Product Configurator for configuration? Refer to Jira detailed in Product Configurator specifications - How can we evaluate Customer Credit (refer to Function 205 for definition): can this check be added in TMF696 - Risk Management API? - AP-6002 - TMF696 - create real Task resource(s) (BACKLOG) - How can we manage/represent mass operations in general, and mass transaction ordering in particular? - [AP-6617] TMF622 - specialize ProductOrderItem for ProductSpecification and ProductOffering - TM Forum Jira

5.3. Further resources

This component is involved in use cases described in IG1228 How to use ODA - Using Open APIs to realize Use Cases.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	29-Mar-2022	Gaetano Biancardi, Anastasios Sarantis, Sylvie Demarest, Emmanuel A. Otchere	First version of document
1.0.1	25-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
2.0.1	19-Aug-2024	Sylvie Demarest	Updated Component Template; Jira reference AP-6617 added to section 5.2.4 as requested by Olivier Arnaud; final edits prior to publication
2.0.1	06-Sep-2024	Amaia White	Final edits prior to publication
2.1.0	12-Nov-2024	Gaetano Biancardi	API version, only major version to be specified
2.1.0	26-Nov-2024	Amaia White	Final edits prior to publication
2.1.1	08-Jul-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03-Jun-2025; Jira reference AP-6617 added to section 5.2.4 as requested by Olivier Arnaud; final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	29-Mar-2022	Goutham Babu	Initial release
Production	20-May-2022	Adrienne Walcott	Updated to reflect TM Forum Approved status

Pre-production	14-Aug-2023	Amaia White	New release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	06-Sep-2024	Amaia White	New release 2.0.1
Production	01-Nov-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	26-Nov-2024	Amaia White	New release 2.1.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	08-Jul-2025	Rosie Wilson	New release 2.1.1

6.2. Acknowledgements

Team Member	Company	Role
Gaetano Biancardi	Accenture	Reviewer
Anastasios Sarantis	Vodafone	Reviewer
Sylvie Demarest	Orange	Editor
Elisabeth Andersson	MATRIX	Reviewer
Emmanuel A. Otchere	Huawei	Additional Input
Ian Turkington	TM Forum	Additional Input
Hugo Vaughan (TM Forum)	TM Forum	Additional Input
Abinash Vishwakarma	Netcracker	Reviewer
Jose Macaluso	Xacria	Reviewer
Kamal Maghsoudlou	Ericsson	Reviewer
Milind Bhagwat	BT	Reviewer
Ritu Arora	BT	Reviewer



Matteo Destino	Accenture	Reviewer
Rosie Wilson	TM Forum	Additional Input